Supplementary: Including ancestral state estimation uncertainty can improve disparity analyses

Caleb Scutt[1,2*], Natalie Cooper[2], Gavin Thomas[1], Thomas Guillerme[1]

[1]School of Biosciences, University of Sheffield, Sheffield, UK

[2]Science Group, Natural History Museum, London, UK

*Corresponding author: cnscutt1@sheffield.ac.uk

S1 Supplementary Tables

Available on Dryad at `https://doi.org/10.5061/dryad.280gb5n4p`.

Table S1 `/continuous/continuous_aggregated_method_multcomp.csv`

Table S2 `/continuous/continuous_aggregated_method_model_fossil_multcomp.csv`

Table S3 `/discrete/discrete_aggregated_method_multcomp.csv`

Table S4 `/discrete/discrete_aggregated_method_model_fossil_multcomp.csv`

S2 Supplementary Figures

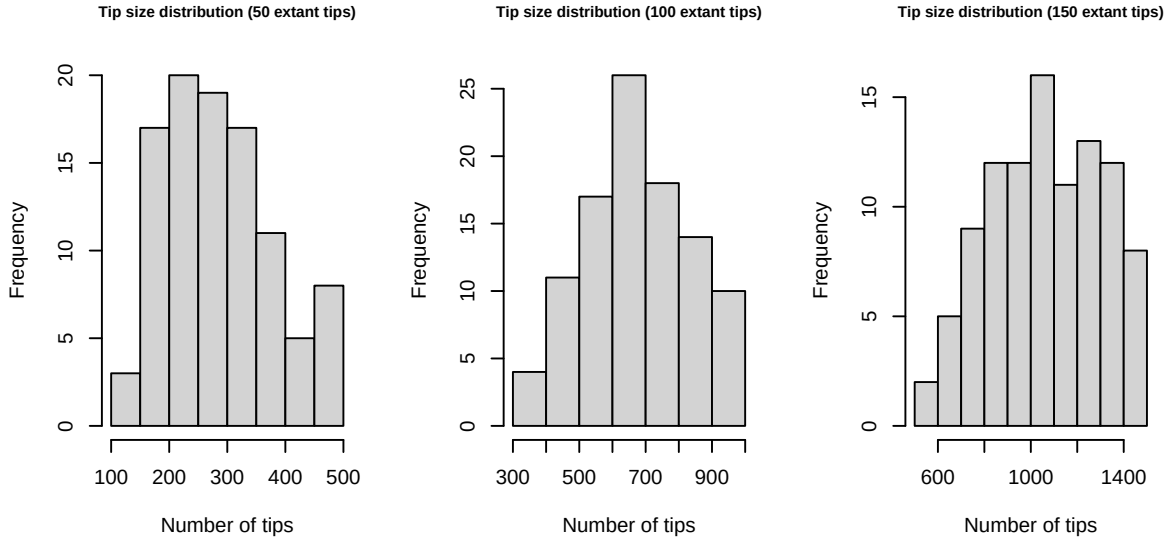

Figure S1: **Distribution of simulated tree sizes.** Histograms show the number of total tips across the simulated trees targeted to have 50, 100 or 150 extant tips. Minor variations around the target sizes reflect the stochastic nature of the birth-death simulation process.

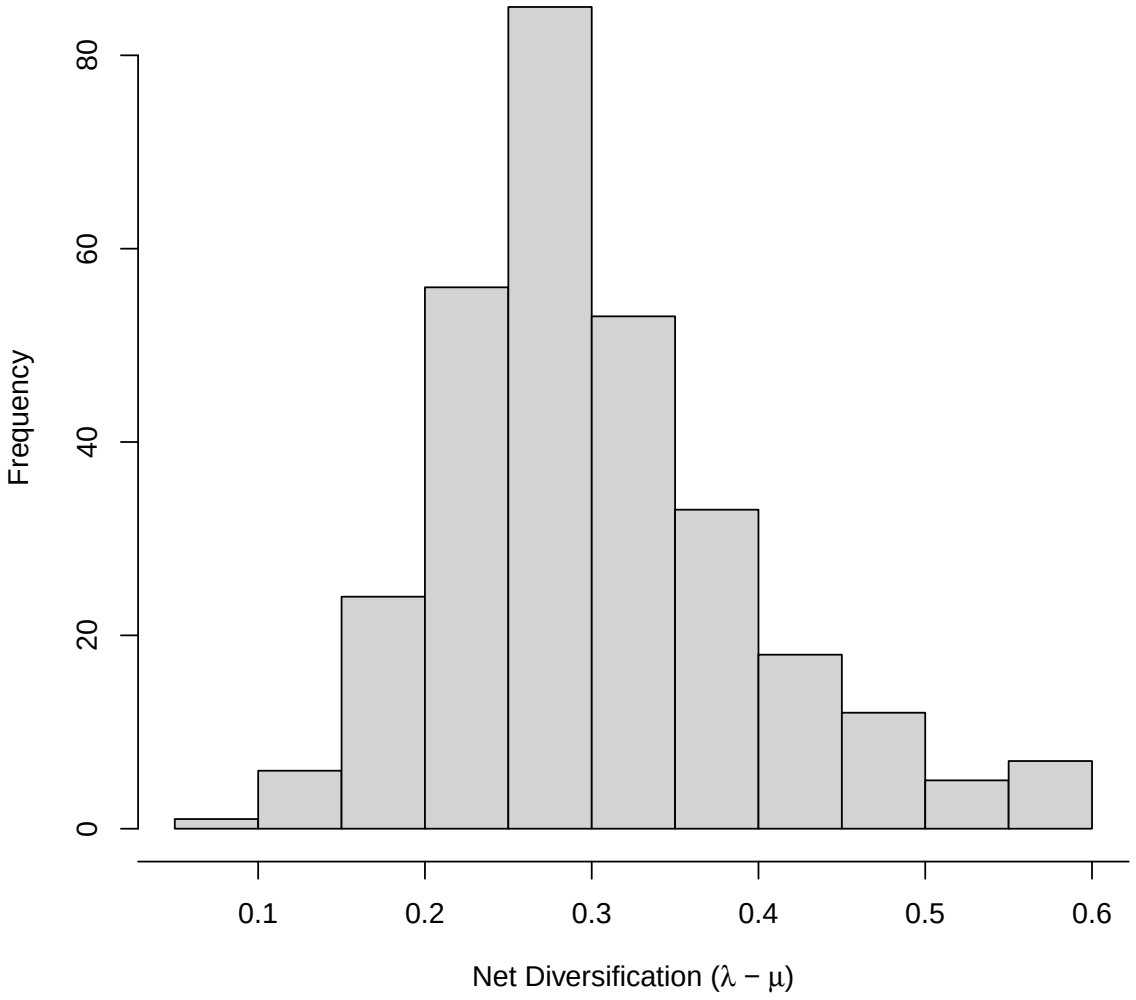

Figure S2: **Net diversification across simulated trees.** Distribution of realised net diversification (speciation minus extinction) values across all simulated phylogenies.

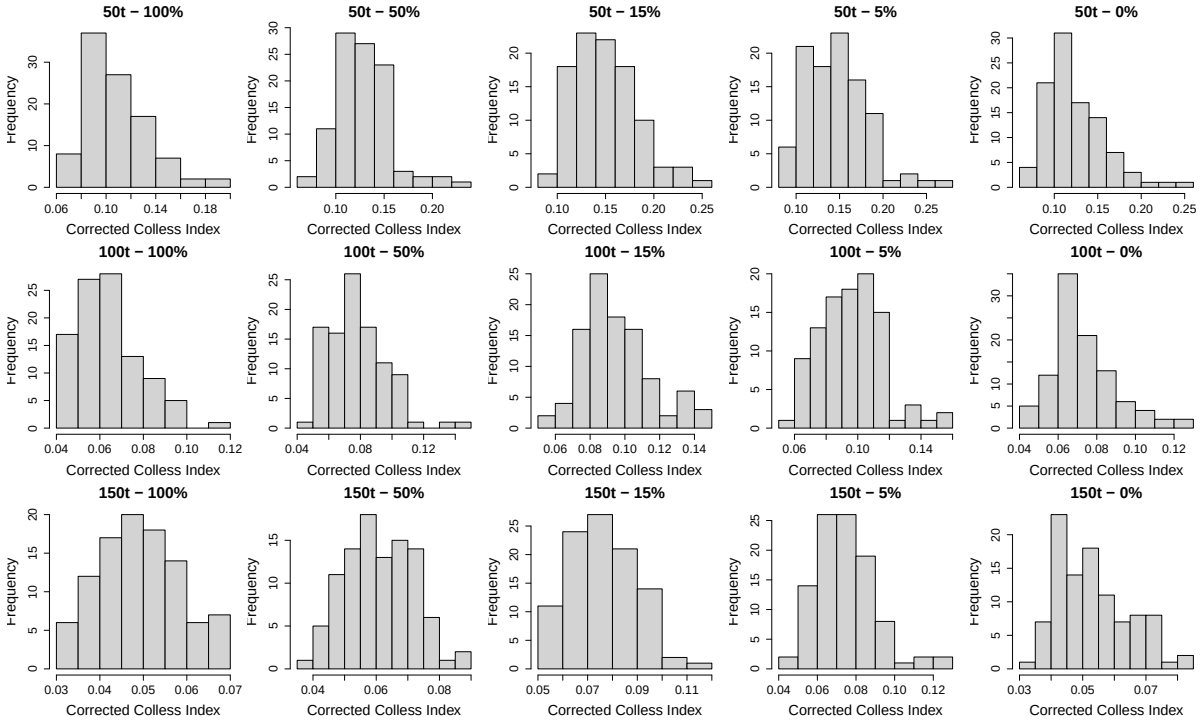

Figure S3: **Main analyses tree symmetry distributions.** Distribution of tree symmetry values measured across the trees used in the main analysis using the corrected Colless index (Heard, 1992). Lower values indicate more symmetrical trees.

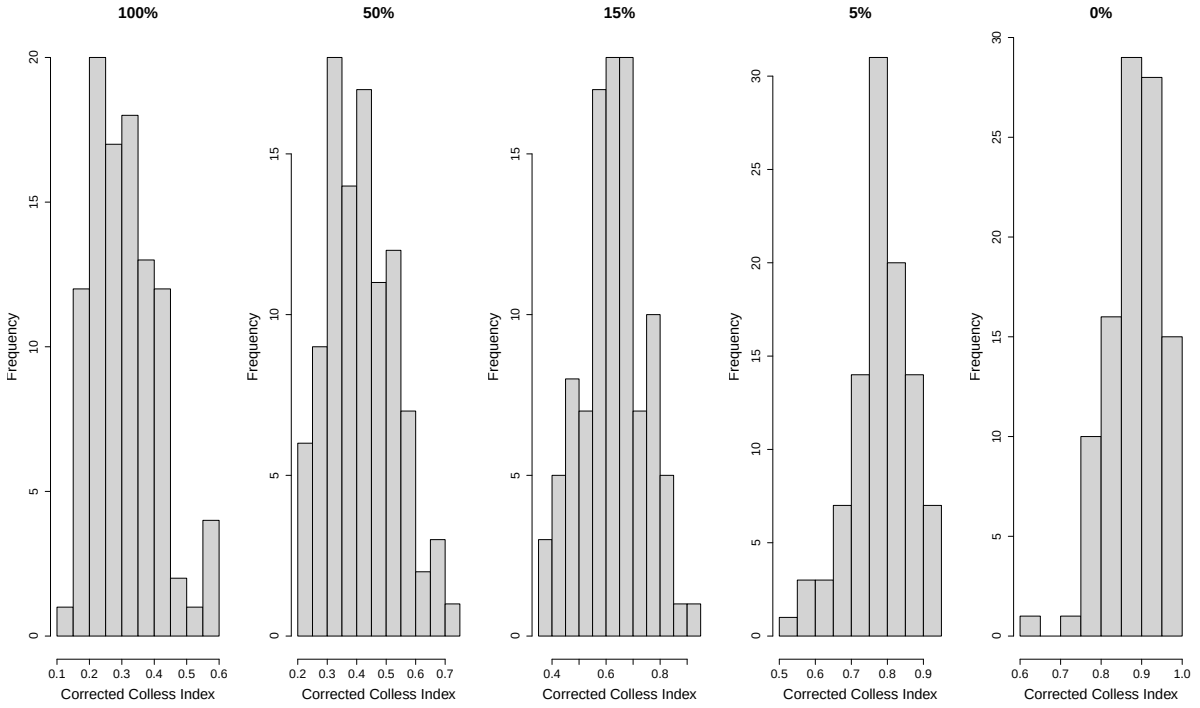

Figure S4: **Asymmetrical trees sensitivity test Colless index distributions.** Distribution of tree symmetry values measured across the trees used in the asymmetry sensitivity test using the corrected Colless index (Heard, 1992).

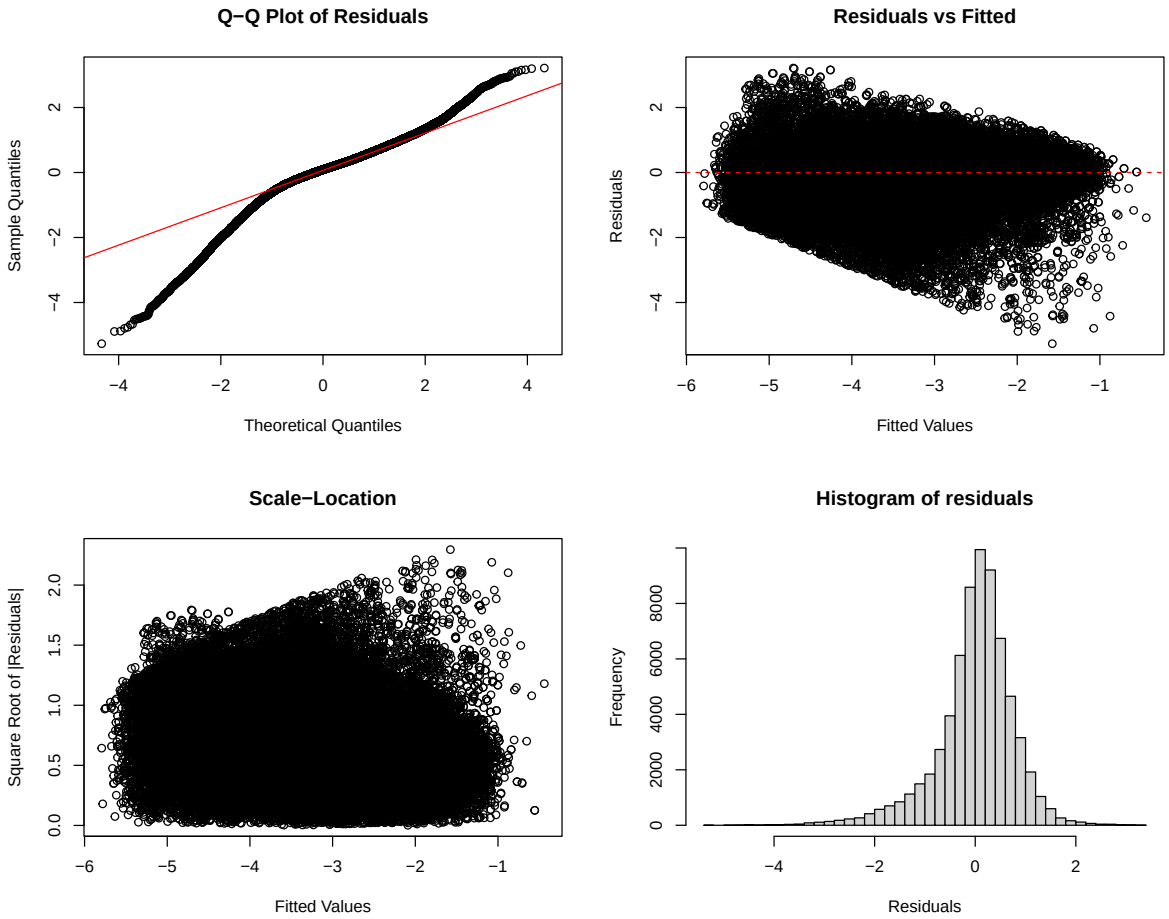

Figure S5: **Diagnostic plots for the aggregated continuous trait linear mixed-effects model**. Residual diagnostic plots used to assess the fit of the linear mixed-effects model fitted to the aggregated continuous trait results.

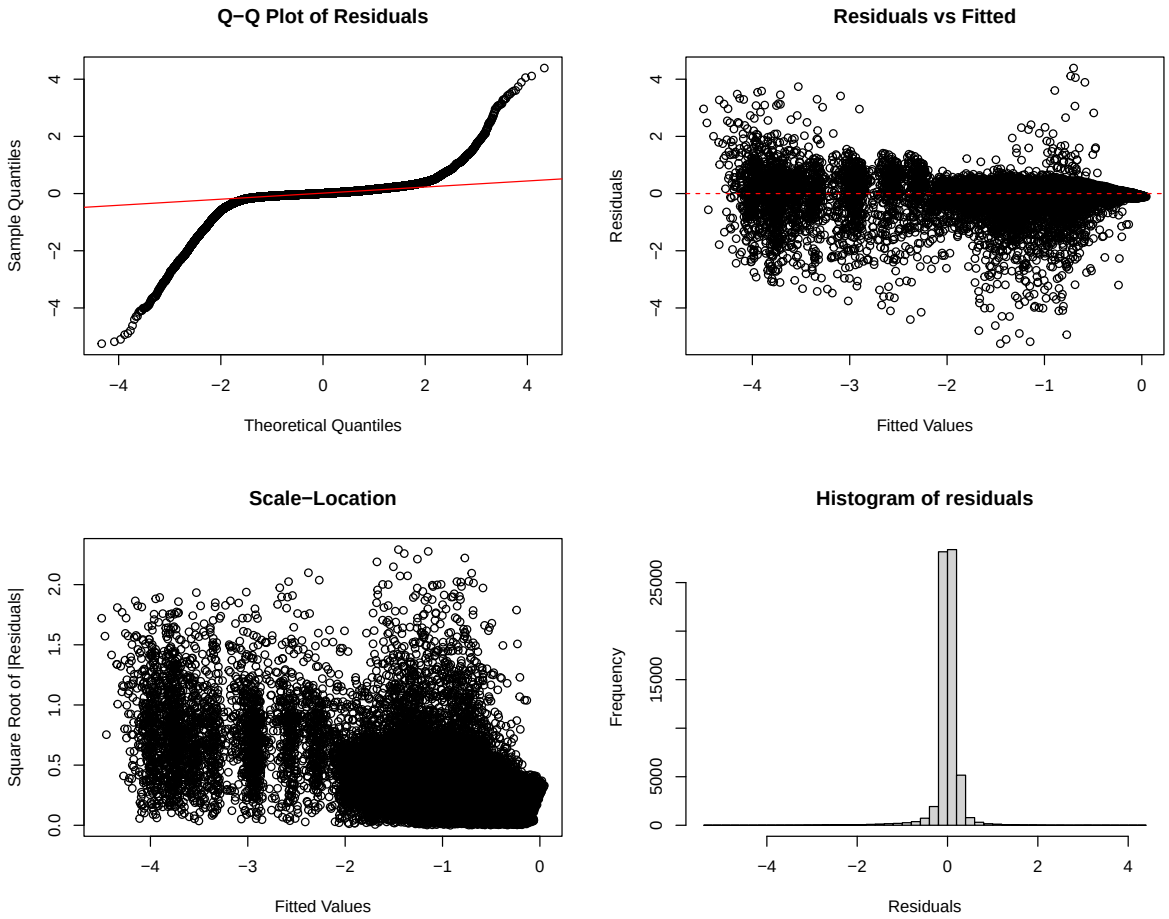

Figure S6: **Diagnostic plots for the aggregated discrete trait linear mixed-effects model** Residual diagnostic plots used to assess the fit of the linear mixed-effects model fitted to the aggregated discrete trait results.

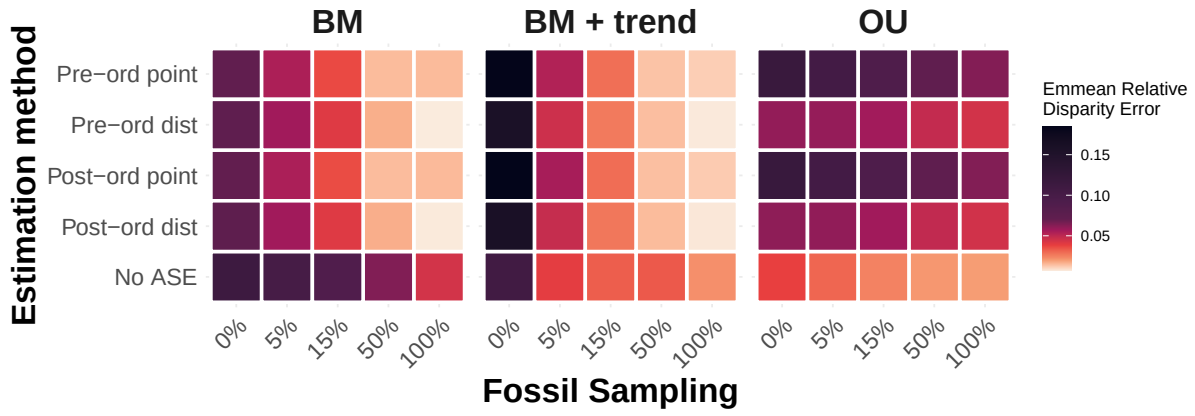

Figure S7: **Weighted LMM continuous heatmap**. Heatmap of estimated marginal mean relative disparity errors from the weighted linear mixed-effects models for continuous traits. Values are averaged across disparity metrics and shown for each combination of estimation method, evolutionary model, and fossil sampling density.

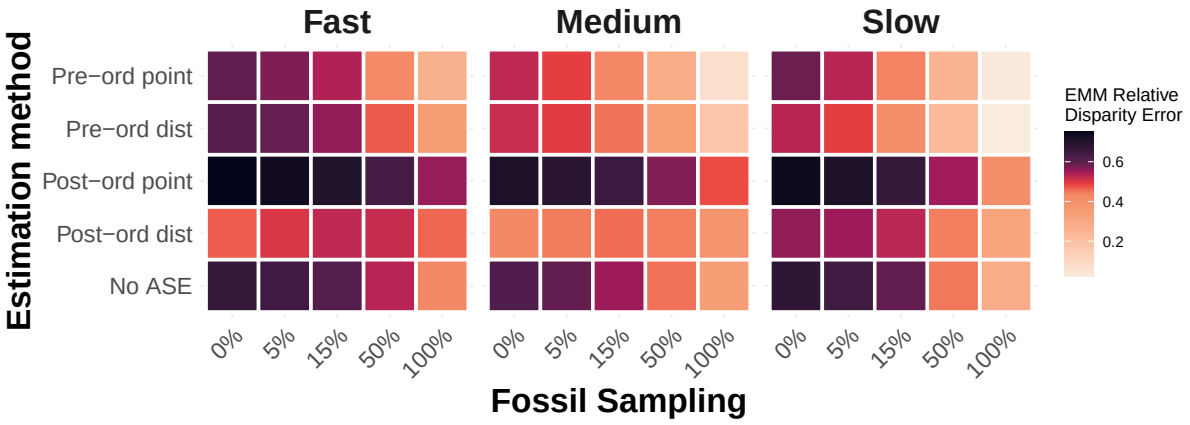

Figure S8: **Weighted LMM discrete heatmap**. Heatmap of estimated marginal mean relative disparity errors from the weighted linear mixed-effects models for discrete traits. Values are averaged across disparity metrics and shown for each combination of estimation method, transition rate, and fossil sampling density.

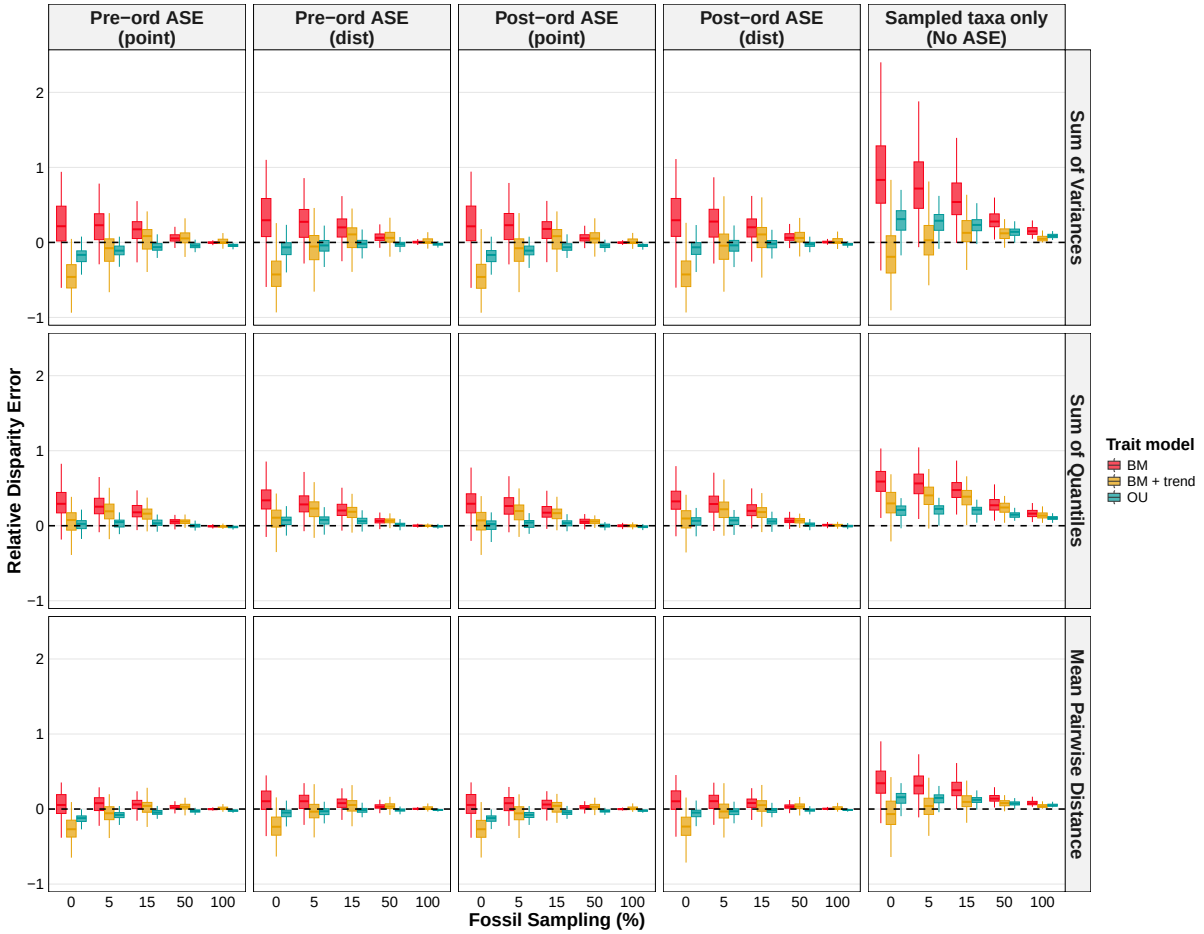

Figure S9: **Continuous trait disparity error under asymmetrical trees**. Relative disparity error returned from pre-ordination point ASE (ancestral state estimation), pre-ordination distribution ASE, post-ordination point ASE, post-ordination distribution ASE and no ASE. These disparity errors were calculated across three evolutionary models (Brownian motion [BM], BM+trend and Ornstein-Uhlenbeck [OU]), five simulated fossil sampling densities (100%, 50%, 15%, 5% and 0%) using the tip-only scheme and three disparity metrics (Sum of variances, sum of quantiles and mean pairwise distance). Values closest to 0 (dashed horizontal line) were most accurate; negative errors are under-estimates and positive are over-estimates.

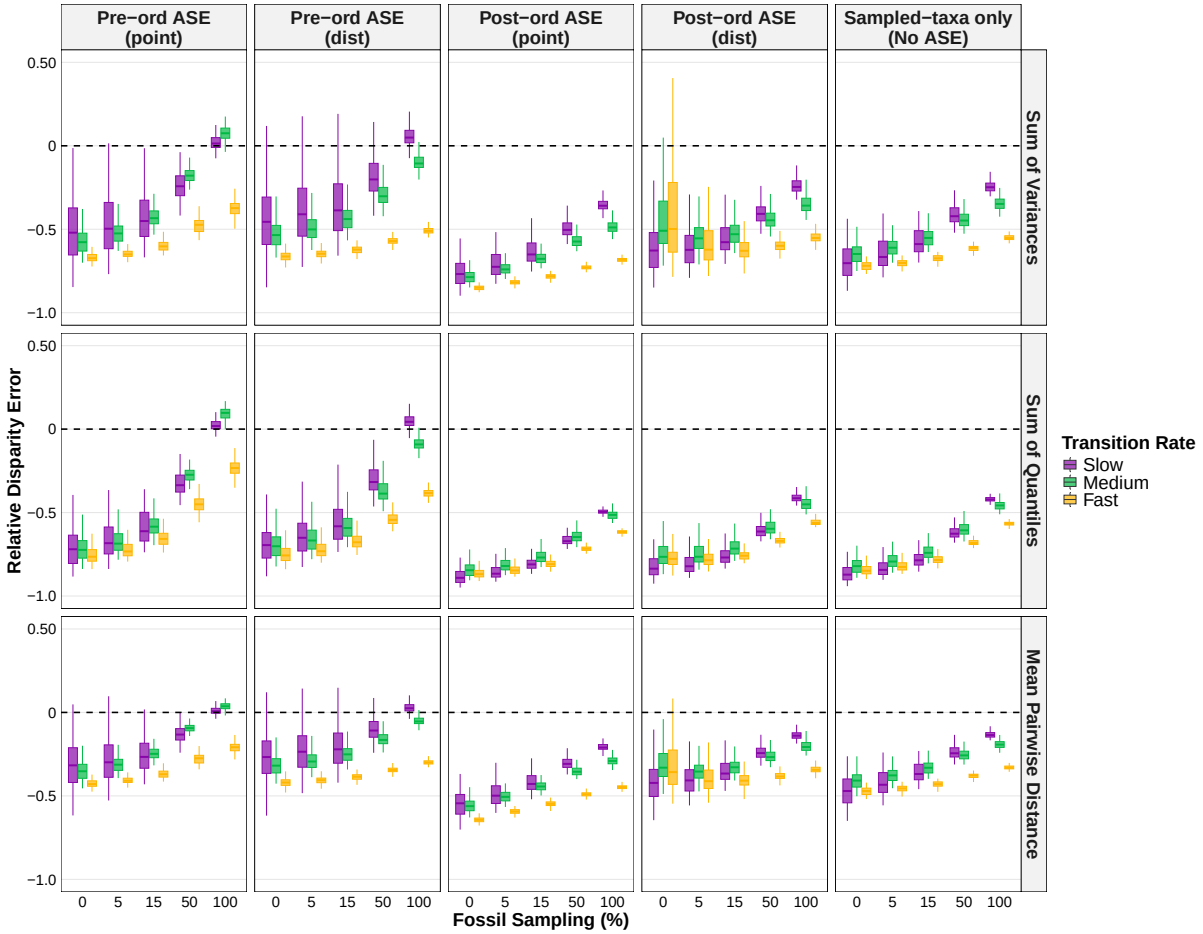

Figure S10: **Discrete trait disparity error under asymmetrical trees**. Relative disparity error across ancestral state estimation methods and fossil sampling densities for discrete traits simulated on asymmetrical phylogenies. Results are shown for the tip-only fossil sampling scheme and are calculated across the transition rates and disparity metrics used in the main analysis. Values closer to zero indicate more accurate recovery of the true disparity. Negative values indicate underestimation and positive values indicate overestimation.

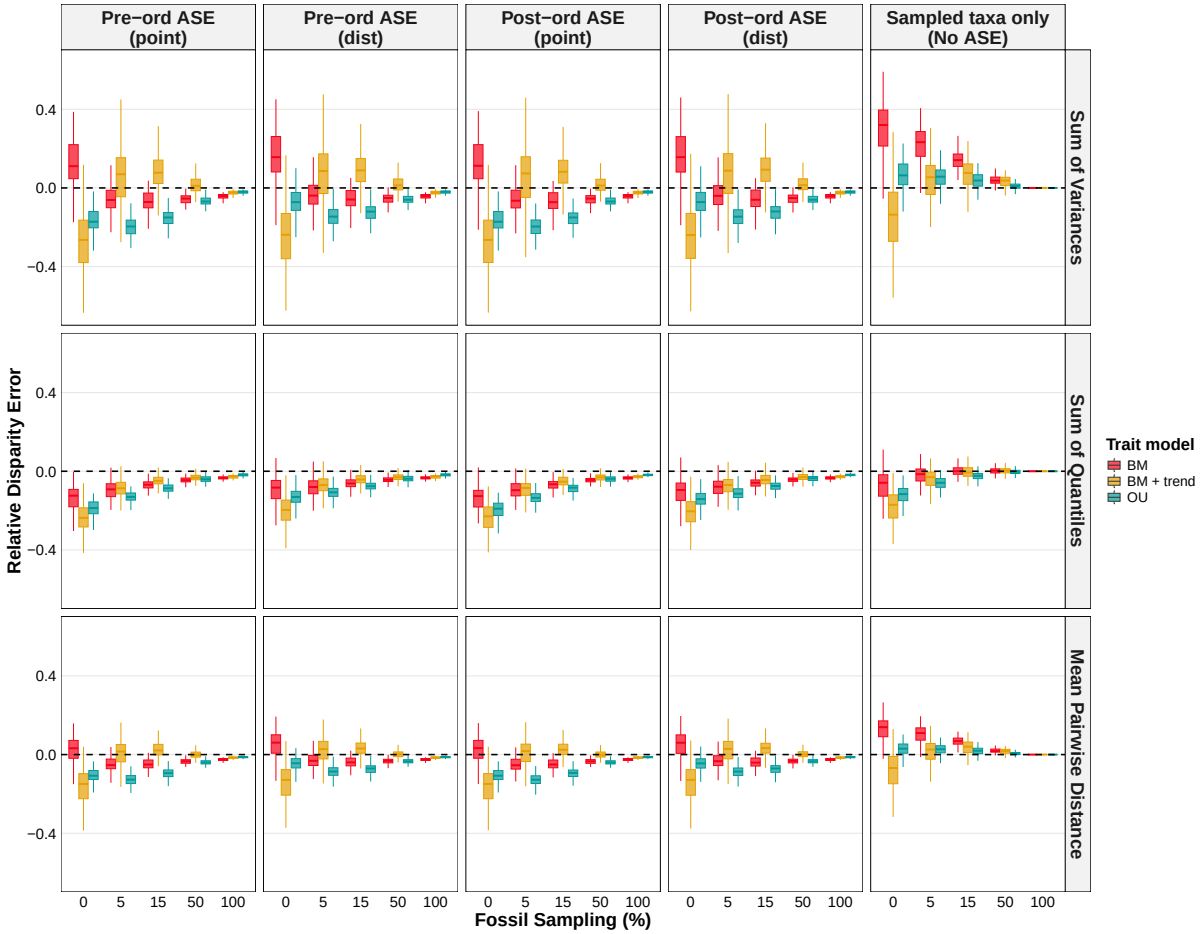

Figure S11: **Continuous disparity error across estimation methods using the alternative tip+nodes fossil sampling scheme**. Relative disparity error returned from pre-ordination point ASE (ancestral state estimation), pre-ordination distribution ASE, post-ordination point ASE, post-ordination distribution ASE and no ASE. These disparity errors were calculated across three evolutionary models (Brownian motion [BM], BM+trend and Ornstein-Uhlenbeck [OU]), five simulated fossil sampling densities (100%, 50%, 15%, 5% and 0%) using the 'tips+nodes' scheme and three disparity metrics (Sum of variances, sum of quantiles and mean pairwise distance). Values closest to 0 (dashed horizontal line) were most accurate; negative errors are under-estimates and positive are over-estimates.

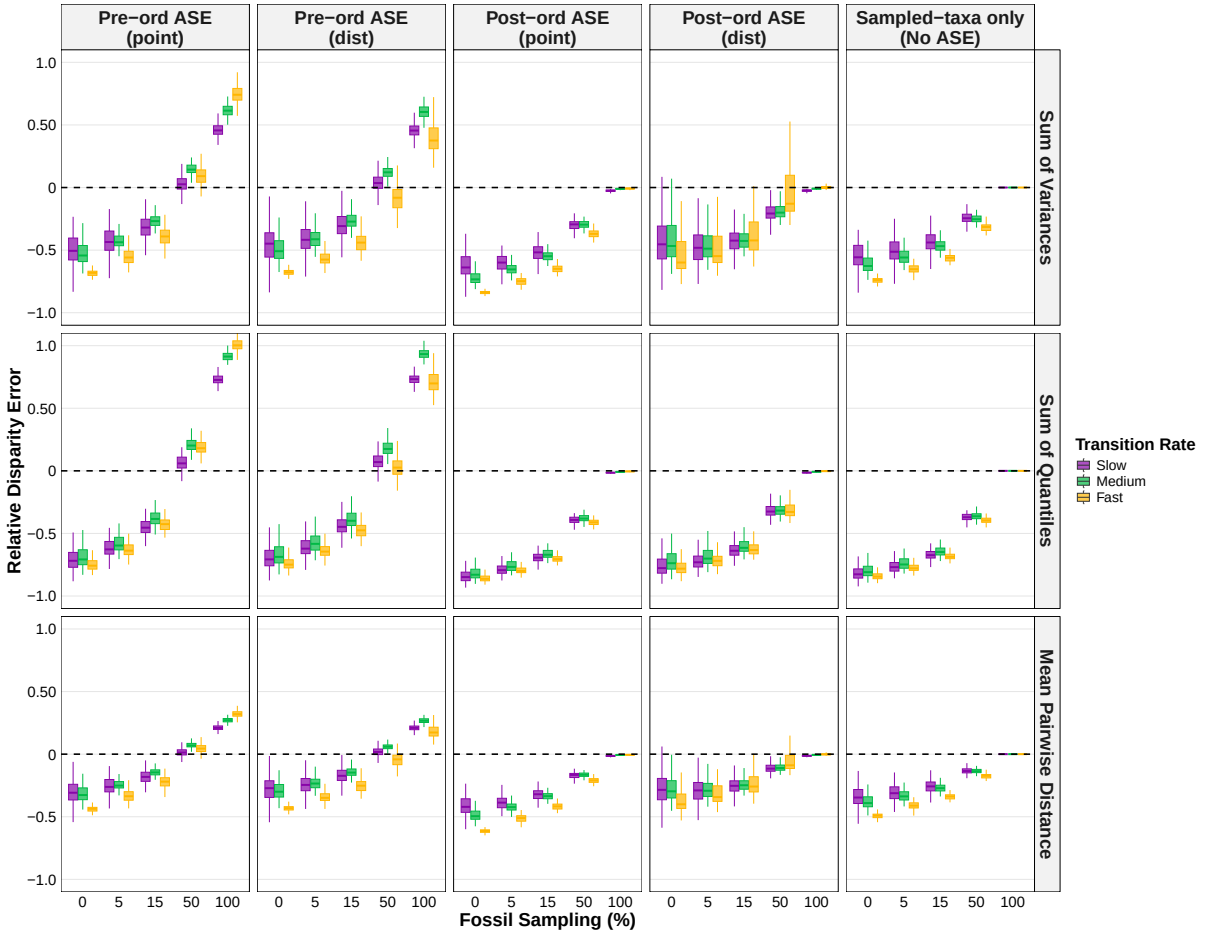

Figure S12: **Discrete disparity error across estimation methods using the alternative tip+nodes fossil sampling scheme**. Relative disparity error returned from pre-ordination point ASE (ancestral state estimation), pre-ordination distribution ASE, post-ordination point ASE, post-ordination distribution ASE and no ASE. These disparity errors were calculated across three transition rates (fast (0.9), medium (0.1) and slow (0.01)), five simulated fossil sampling densities (100%, 50%, 15%, 5% and 0%) under 'tips+nodes' scheme and three disparity metrics (Sum of variances, sum of quantiles and mean pairwise distance). Values closest to 0 (dashed horizontal line) were most accurate; negative errors are under-estimates and positive are over-estimates.

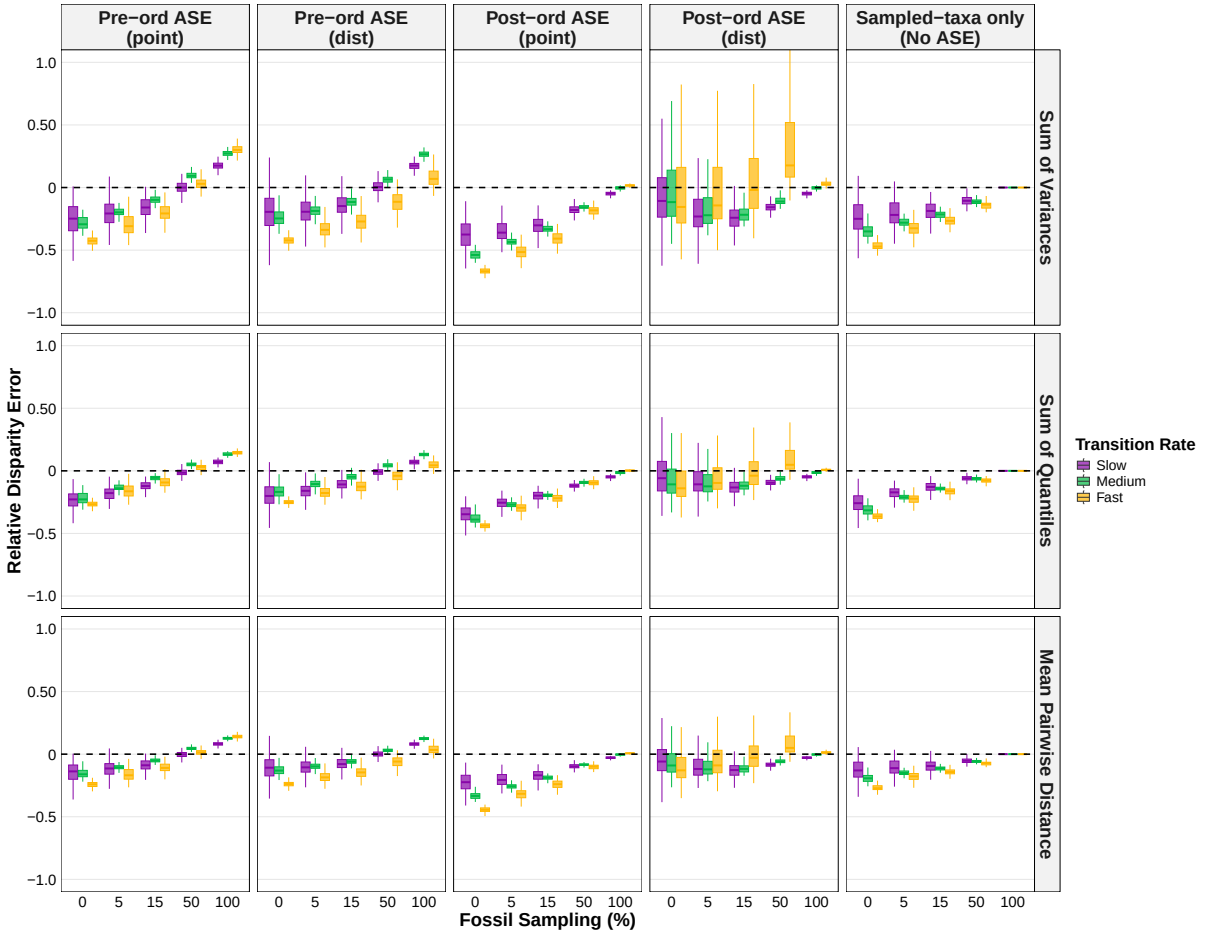

Figure S13: **Discrete disparity error across estimation methods using the alternative tip+nodes fossil sampling scheme with common number of ordination axes across all fossil sampling densities**. Relative disparity error returned from pre-ordination point ASE (ancestral state estimation), pre-ordination distribution ASE, post-ordination point ASE, post-ordination distribution ASE and no ASE. These disparity errors were calculated across three transition rates (fast (0.9), medium (0.1) and slow (0.01)), five simulated fossil sampling densities (100%, 50%, 15%, 5% and 0%) under the 'tips+nodes' scheme and three disparity metrics (sum of variances, sum of quantiles and mean pairwise distance). Values closest to 0 (dashed horizontal line) were most accurate; negative errors are under-estimates and positive are over-estimates.

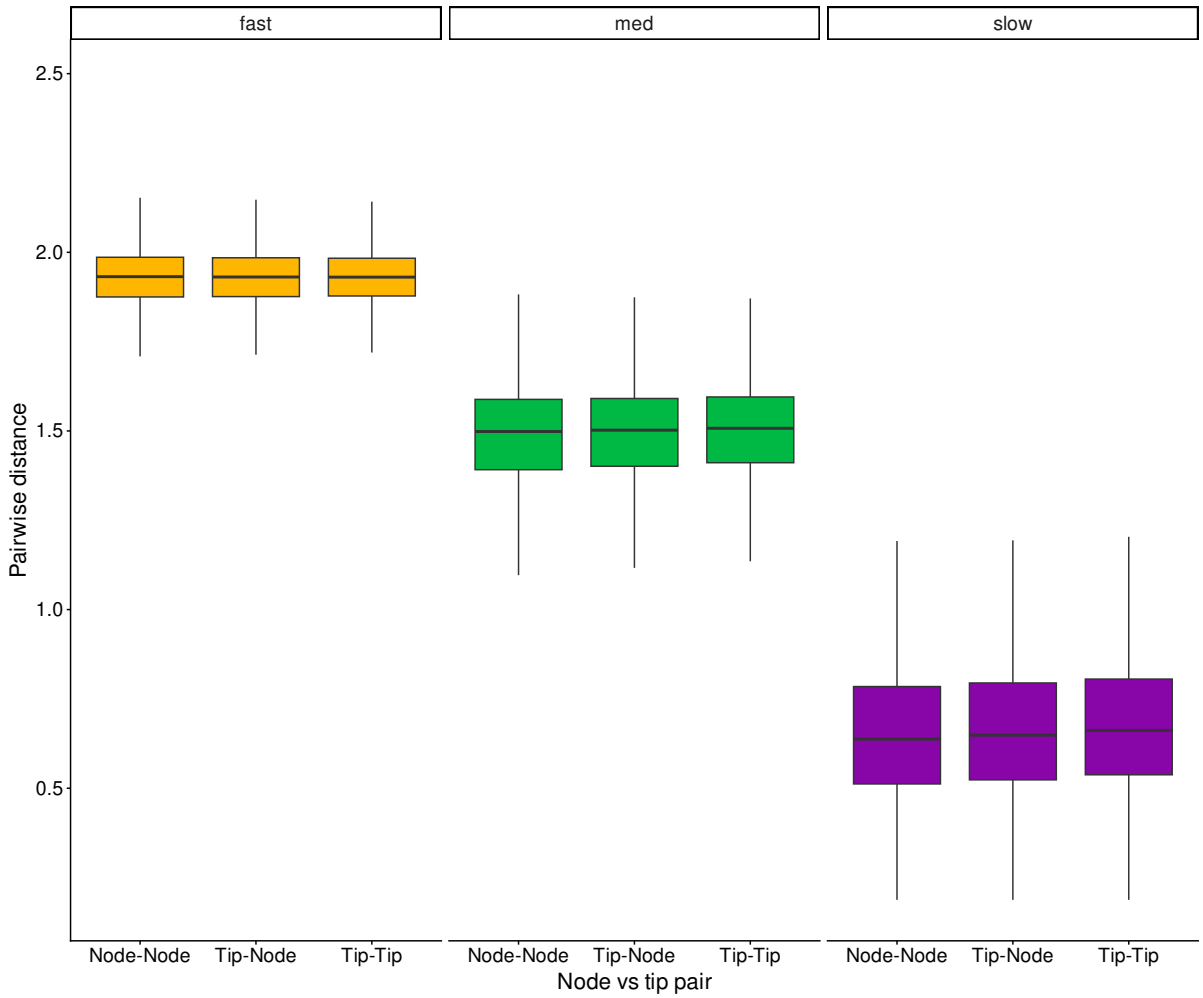

Figure S14: Pairwise Euclidean distances between tips and nodes in true ordinations of discrete traits. Distances calculated across all axes from the 100 true discrete trait ordinations from 50-living taxa subsample.

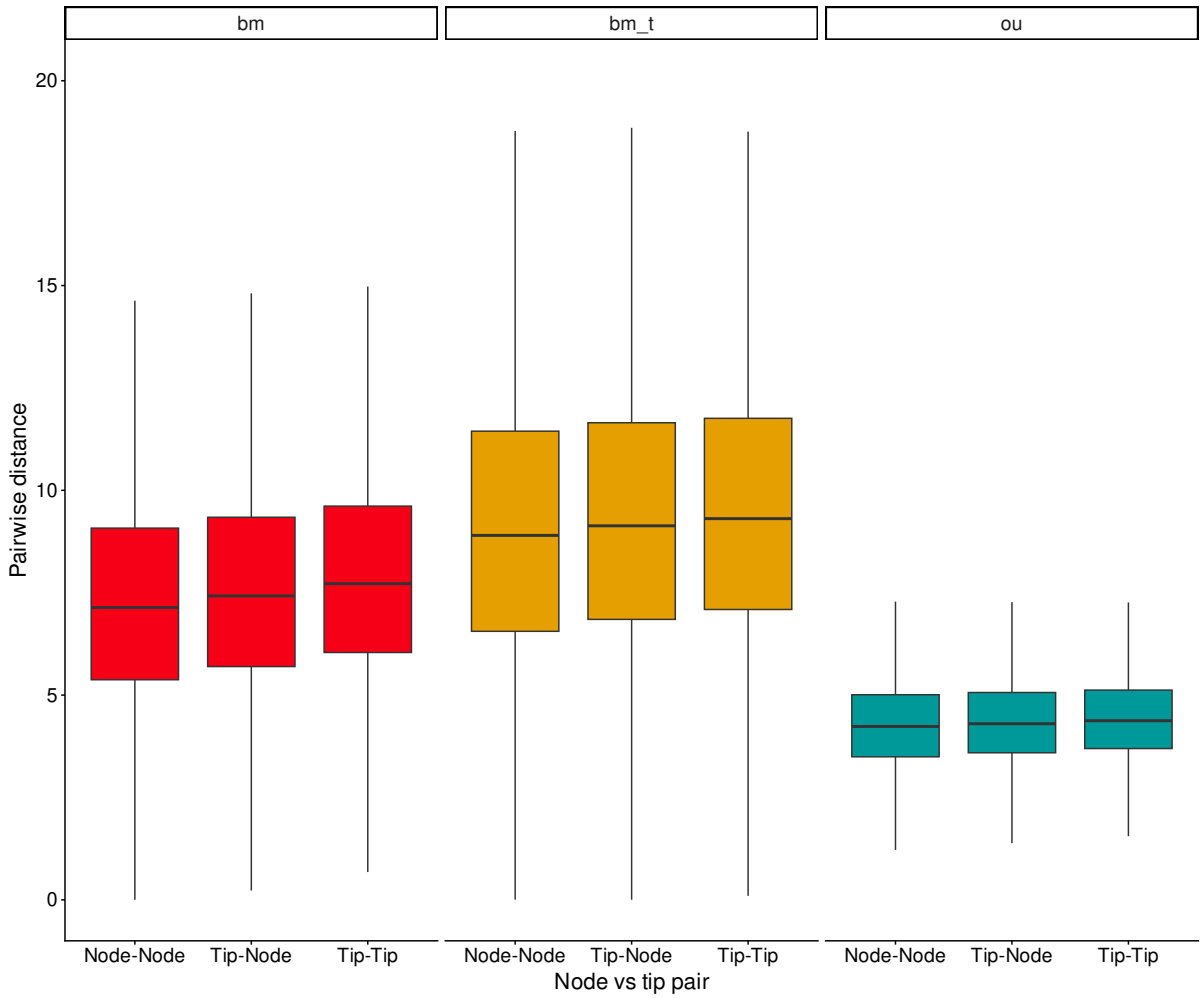

Figure S15: Pairwise Euclidean distances between tips and nodes in true ordinations of continuous traits. Distances calculated across all axes from the 100 true continuous ordinations from 50-living taxa subsample.

References

Stephen B Heard. Patterns in tree balance among cladistic, phenetic, and randomly generated phylogenetic trees. *Evolution*, 46(6):1818–1826, 1992.